

Out of Work: Forecasting and Explaining U.S. Unemployment Through Macroeconomic Indicators

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Abstract

Can the U.S. unemployment rate be predicted before it rises? This project used public U.S. economic data to forecast the **national unemployment rate** 3, 6, and 12 months ahead. Results show that short-term forecasts are useful, while longer-term forecasts are less reliable.

Introduction & Motivation

Problem & Question

Economic changes can quickly affect the U.S. unemployment rate, making early prediction valuable for planning. This study asks:

- Can the **U.S. unemployment rate** be forecast **3, 6, and 12 months ahead** using public economic indicators?
- Which economic indicators have the **strongest historical relationship** with unemployment?

Impact & Stakeholders

The results can support **policymakers, economists, businesses, investors, and workforce agencies** in planning for future labor market changes.

Background & Related Work

- Broad economic indicators can improve macroeconomic forecasting [1], [2].
- Consumer sentiment may provide information about future economic activity [3].
- **This study compares 3-, 6-, and 12-month forecasts** and examines **historical relationships between public economic indicators and unemployment**.

Data, Engineering & Ethics

Source & Scale:

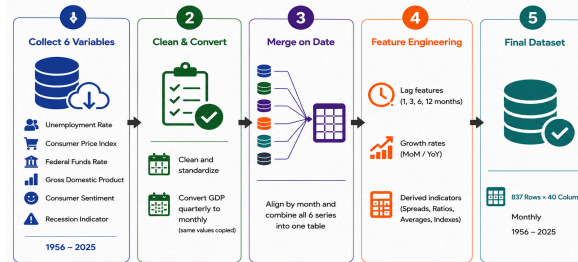
Six public U.S. economic series produced **837 monthly records** from **April 1956 to December 2025**.

Pipeline:

Download Raw Data → Clean → Merge by Date → Feature Engineering → Final Dataset

Ethics:

The data are public and aggregated, with **no personal information**. Results should support planning, not replace human judgment.



Methods & Evaluation

Approach:

Compared **Linear Regression, Ridge Regression, Random Forest, Extra Trees, Gradient Boosting, and XGBoost** to evaluate **linear and nonlinear forecasting** across 3-, 6-, and 12-month horizons.

Validation:

Models were selected by **validation RMSE**; tested using **MAE, RMSE, and R^2** .

Evidence:

Compared against **persistence** and **unemployment-only** baselines to verify performance gains.

Results & Visualizations

Figure 1. Actual vs. predicted unemployment rate for the 3-month horizon using Ridge regression, Test set: $n = 321$ months

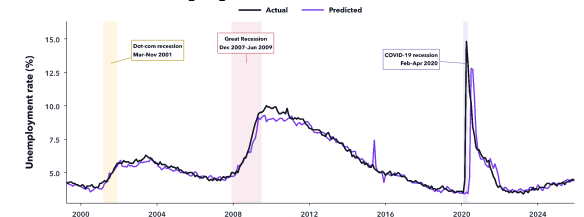
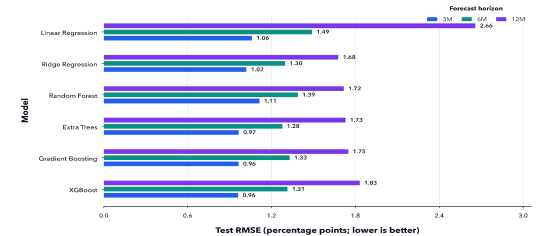


Figure 2. Forecast Accuracy of Six Machine Learning Models



Discussion, Limitations & Conclusions

Discussion:

- **3-month forecasts** provided the strongest predictive performance; **6- and 12-month** accuracy declined.

Limitations & future work:

- The model uses **six national indicators** and does not capture **state- or industry-level** labor markets. Future work should include **finer-grained data** and evaluate additional forecasting methods.

Conclusions:

- **Short-term unemployment forecasting is practical**. **Consumer sentiment** showed the **strongest overall historical relationship with unemployment**, followed by **recession status**. These indicators can support earlier economic planning.

Reproducibility & References

Repository: <https://github.com/manishkallu01-wq/DATA510-Session-2> · Key dependencies: Python 3.13, pandas, NumPy, scikit-learn, matplotlib

References: [1] Stock & Watson (2002), *Macroeconomic Forecasting Using Diffusion Indexes* · [2] McCracken & Ng (2016), *FRED-MD: A Monthly Database for Macroeconomic Research*
[3] Barsky & Sims (2012), *Consumer Confidence and Future Economic Activity*

